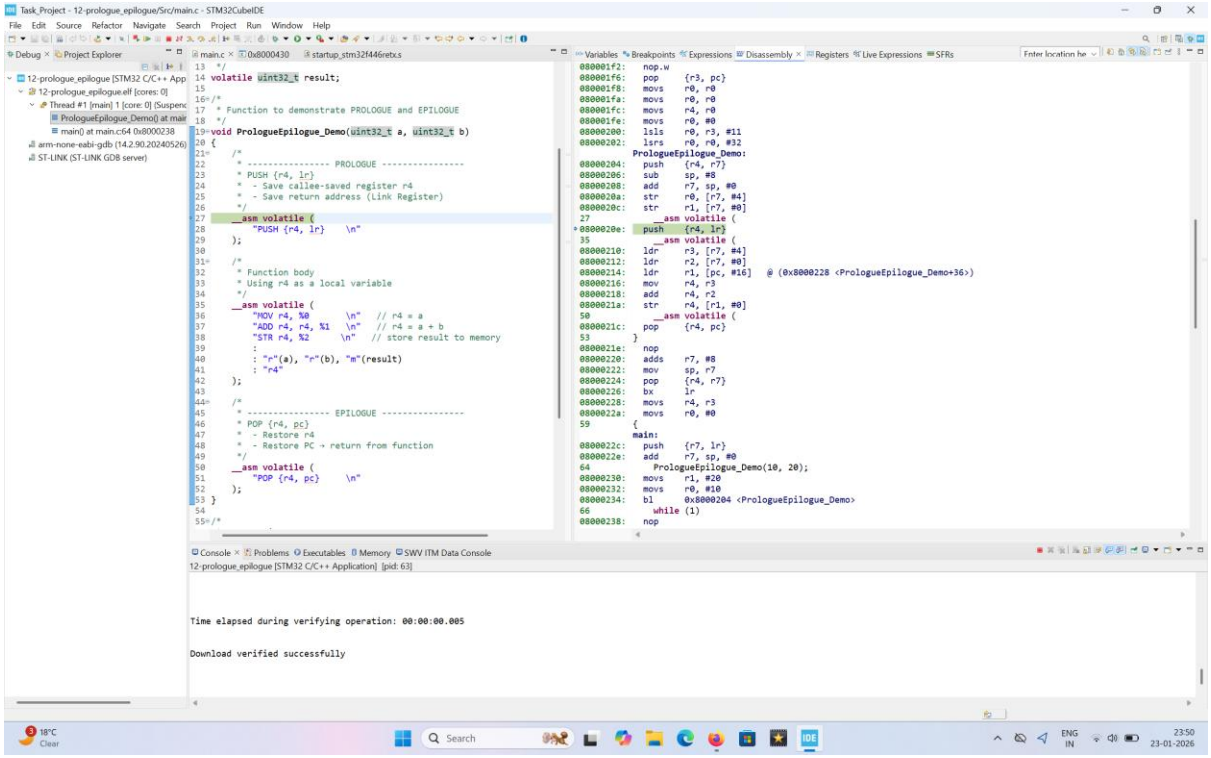
Extra Task prologue and Epilogue

```
/**

*********************************************************************
*******
 * @file           : main.c
 * @author         : Sagar More
 * @brief          : Prologue and Epilogue

*********************************************************************
*******
 */

#include <stdint.h>

/*
 * Dummy global variable so compiler does not optimize everything
 */
volatile uint32_t result;

/*
 * Function to demonstrate PROLOGUE and EPILOGUE
 */
void PrologueEpilogue_Demo(uint32_t a, uint32_t b)
{
    /*
     * ---------------- PROLOGUE ----------------
     * PUSH {r4, lr}
     *  - Save callee-saved register r4
     *  - Save return address (Link Register)
     */
    __asm volatile (
        "PUSH {r4, lr}    \n"
    );

    /*
     * Function body
     * Using r4 as a local variable
     */
    __asm volatile (
        "MOV r4, %0       \n"   // r4 = a
        "ADD r4, r4, %1   \n"   // r4 = a + b
        "STR r4, %2      \n"   // store result to memory
        :
        : "r"(a), "r"(b), "m"(result)
        : "r4"
    );

    /*
     * ---------------- EPILOGUE ----------------
     * POP {r4, pc}
     *  - Restore r4
     *  - Restore PC → return from function
```

```
     */
    __asm volatile (
        "POP {r4, pc}     \n"
    );
}

/*
 * Entry point
 */
int main(void)
{
    /*
     * Call function
     * Parameters passed via registers (AAPCS)
     */
    PrologueEpilogue_Demo(10, 20);

    while (1)
    {
        /* Infinite loop */
    }
}
```

Output Images


Task_Project - 12-prologue_epilogue/Src/main.c - STM32CubeIDE
File Edit Source Refactor Navigate Search Project Run Window Help
Debug
Project Explorer
12-prologue_epilogue [STM32 C/C++ App
12-prologue_epilogue.elf [cores: 0]
Thread #1 [main] 1 [core: 0] (Suspenc
PrologueEpilogue_Demo() at mair
main() at main.c64 0x8000238
arm-none-eabi-gdb (14.2.90.20240526)
ST-LINK (ST-LINK GDB server)
main.c
0x8000430
startup_stm32f446retx.s
13 */
14 volatile uint32_t result;
15
16 /*
17 * Function to demonstrate PROLOGUE and EPILOGUE
18 */
19 void PrologueEpilogue_Demo(uint32_t a, uint32_t b)
20 {
21 /*
22 * ---------- PROLOGUE ----------
23 * PUSH {r4, lr}
24 * - Save callee-saved register r4
25 * - Save return address (Link Register)
26 */
27 __asm volatile (
28 "PUSH {r4, lr} \n"
29);
30
31 /*
32 * Function body
33 * Using r4 as a local variable
34 */
35 __asm volatile (
36 "MOV r4, %0 \n" // r4 = a
37 "ADD r4, r4, %1 \n" // r4 = a + b
38 "STR r4, %2 \n" // store result to memory
39 :
40 : "r"(a), "r"(b), "m"(result)
41 : "r4"
42);
43
44 /*
45 * ---------- EPILOGUE ----------
46 * POP {r4, pc}
47 * - Restore r4
48 * - Restore PC → return from function
49 */
50 __asm volatile (
51 "POP {r4, pc} \n"
52);
53 }
54
55 /*
Variables
Breakpoints
Expressions
Disassembly
Registers
Live Expressions
SFRs
080001f2: nop.w
080001f6: pop {r3, pc}
080001f8: movs r0, r0
080001fa: movs r0, r0
080001fc: movs r4, r0
080001fe: movs r0, #0
08000200: lsls r0, r3, #11
08000202: lsrs r0, r0, #32
PrologueEpilogue_Demo:
08000204: push {r4, r7}
08000206: sub sp, #8
08000208: add r7, sp, #0
0800020a: str r0, [r7, #4]
0800020c: str r1, [r7, #0]
27 __asm volatile (
0800020e: push {r4, lr}
35 __asm volatile (
08000210: ldr r3, [r7, #4]
08000212: ldr r2, [r7, #0]
08000214: ldr r1, [pc, #16] @ (0x8000228 <PrologueEpilogue_Demo+36>)
08000216: mov r4, r3
08000218: add r4, r2
0800021a: str r4, [r1, #0]
50 __asm volatile (
0800021c: pop {r4, pc}
53 }
0800021e: nop
08000220: adds r7, #8
08000222: mov sp, r7
08000224: pop {r4, r7}
08000226: bx lr
08000228: movs r4, r3
0800022a: movs r0, #0
59 {
main:
0800022c: push {r7, lr}
0800022e: add r7, sp, #0
64 PrologueEpilogue_Demo(10, 20);
08000230: movs r1, #20
08000232: movs r0, #10
08000234: bl 0x8000204 <PrologueEpilogue_Demo>
66 while (1)
08000238: nop
Console
Problems
Executables
Memory
SWV ITM Data Console
12-prologue_epilogue [STM32 C/C++ Application] [pid: 63]
Time elapsed during verifying operation: 00:00:00.005
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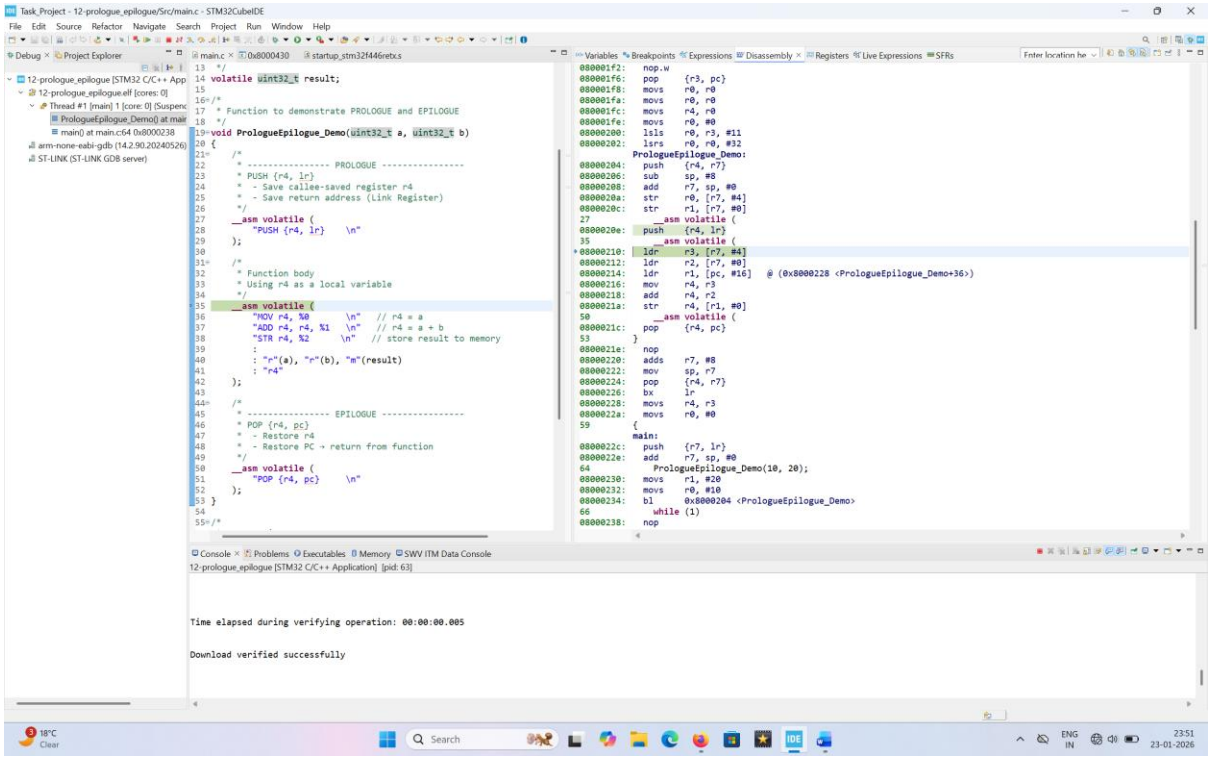
18°C
Clear
Search
ENG
IN
23:50
23-01-2026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32 * Function body
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44 /*
45 * ---------- EPILOGUE ----------
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47 * - Restore r4
48 * - Restore PC → return from function
49 */
50 __asm volatile (
51 "POP {r4, pc} \n"
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53 }
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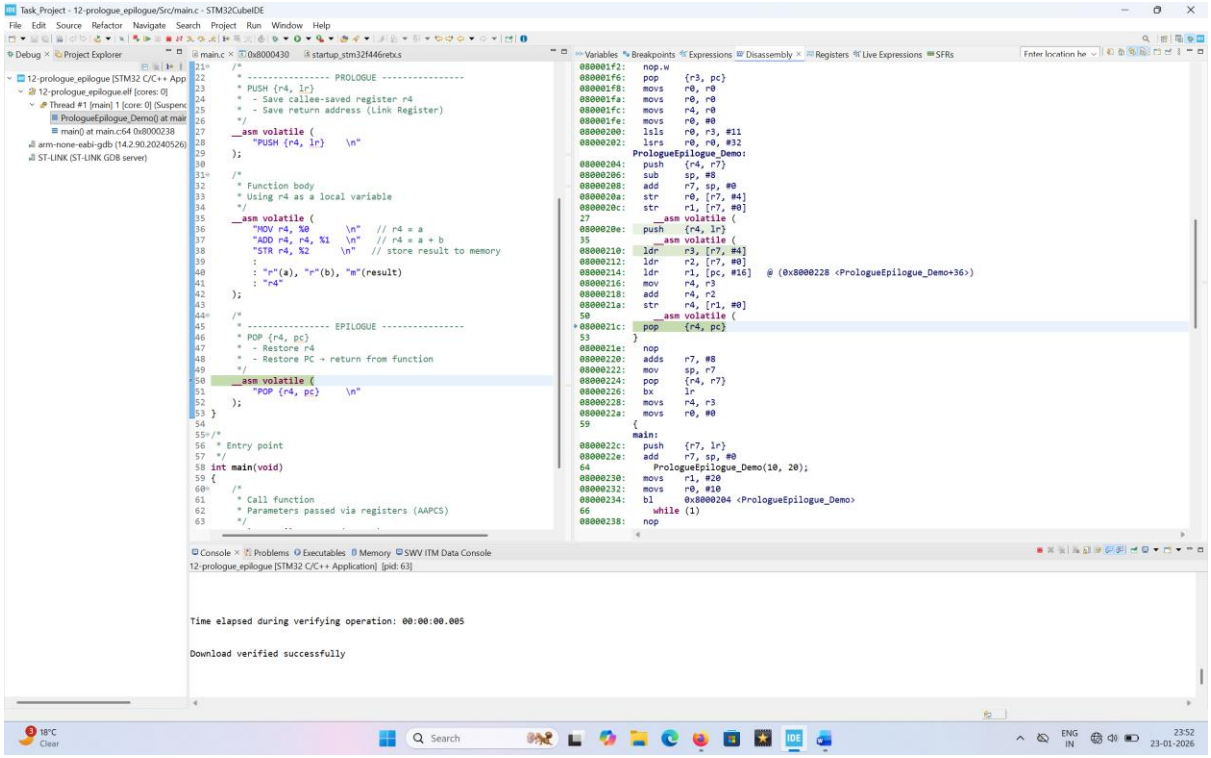
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